Cross-Cutting Exposure as an Engine of Radicalization: Platform Design, Attention, and Geography in a Stochastic Multiplex Model of Opinion Dynamics

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Abstract

Do social-media platforms polarize because they hide disagreement, or because they expose people to it? We study this question in a continuous-time model in which agents diffuse through physical space by Brownian motion while interacting through an adaptive, algorithmically curated digital network, under a finite attention budget that forces digital exposure to displace physical interaction. When social influence is purely assimilative (bounded confidence), the conventional picture holds: opinion-blind long-range exposure heals geographic fragmentation, and algorithmic homophily builds echo chambers. But once influence includes an empirically supported repulsive response to strongly opposed views, the ranking of platform designs inverts: the *neutral*, uncurated platform drives the population to maximal polarization with opinions pinned at the extremes, a controversy-seeking engagement algorithm is nearly as radicalizing, and strong algorithmic homophily—the standard culprit in the echo-chamber narrative—paradoxically shields the population from extremism. We derive this inversion analytically from a two-bloc reduction of the dynamics: the stationary fraction of attention slots pointing across blocs, set by the platform's engagement kernel, determines the radicalization rate, its arrest, and a parameter-free finite-horizon crossover $\lambda_c(T)$ in the digital attention share that reproduces the simulated onset for all three platforms—radicalization is rate-limited, not threshold-limited. Simulations across repulsion parameters, engagement kernels, and system sizes up to $N = 1600$ show that uncurated exposure is the upper envelope of radicalization, with curated platforms approaching that envelope exactly as their curation stops withholding repulsive-zone content. Two further results complete the picture. First, a phase diagram in the mobility–attention plane shows that once the platform owns most of an agent's attention, physical mobility becomes irrelevant to the outcome. Second, under opinion-independent mobility geography never becomes a predictor of opinion—a null result we explain by symmetry—and a Schelling-type homophilic drift of strength χ restores geographic opinion structure only above a Péclet-number threshold $\chi\ell/D \sim 1$. The model offers a mechanistic account of field experiments in which cross-cutting exposure increased polarization, and implies that interventions promoting exposure diversity can backfire precisely when negative influence dominates.

1 Introduction

The most common mechanistic story about platforms and polarization runs through similarity: recommendation algorithms feed users content they already agree with, creating echo chambers whose members drift apart from the rest of society [1–3]. The story suggests an obvious remedy—expose people to the other side. Yet the best-known field experiment on this intervention found the opposite of the intended effect: Republicans exposed for a month to a stream of liberal content became substantially *more* conservative [4]. Social psychology offers a mechanism for such backfire: influence is not always assimilative, and views far beyond an agent's latitude of acceptance can repel rather than attract [5, 6]. What is missing is a quantitative understanding of how this repulsive channel interacts with the *design* of the exposure algorithm: which platforms radicalize, how fast, and above which level of digital attention.

This paper builds a minimal model in which that question has a sharp answer. Agents carry continuous opinions and interact through two layers simultaneously: a physical layer, where encounters are structured by geography—agents diffuse through a two-dimensional world by Brownian motion and influence each other through a short-range bounded-confidence kernel [7–9]—and a digital layer, a directed attention graph whose links are continuously rewired by a platform according to an *engagement kernel* $E(\Delta)$: the probability that content at opinion distance Δ is shown. A finite attention budget couples the layers: the digital attention share λ displaces physical interaction rather than adding to it. On top of this multiplex substrate [10, 11] we add the two empirically motivated ingredients: an assimilation–indifference–repulsion influence function [5, 6] and heavy-tailed influence strengths.

Our headline result is an inversion. With purely assimilative influence, the model reproduces the standard picture—an opinion-blind platform acts like infinite-range mobility and heals fragmentation, while algorithmic homophily freezes fragmentation into spatially delocalized echo chambers. With the repulsive channel active, every platform radicalizes the population, but the ordering reverses: the neutral platform is the *most* polarizing, the controversy-seeking platform nearly as much, and strong algorithmic homophily the *least*, because hiding disagreement starves the repulsive channel. Uncurated exposure turns out to be the upper envelope of radicalization: curated platforms approach it exactly as their curation stops withholding repulsive-zone content.

The inversion is not a simulation curiosity; it follows from the structure of the model, and we derive it in three steps. A two-bloc reduction shows that the stationary fraction of attention slots pointing across blocs, $p(y) = E(2y)/[E(0)+E(2y)]$ for blocs at $\pm y$, controls the radicalization rate $\dot{y} = 2\lambda\eta\, p(y)\, y$; the shapes of $p(y)$ for the three kernels immediately give the ordering, the pinning of the neutral platform at the boundary, and the kinetic arrest of the controversy platform just short of it. The same equation shows that radicalization has *no* threshold in the attention share—any $\lambda > 0$ radicalizes eventually—and yields a parameter-free finite-horizon crossover $\lambda_c(T)$ that reproduces the simulated onset of all three platforms, spanning two orders of magnitude. And the fast-rewiring, well-mixed limit of the model is a McKean–Vlasov particle system whose integration reproduces the full $\mathrm{Var}(x)$-versus-λ curves of the spatial agent-based model.

Two secondary results delimit the role of geography. First, in a phase diagram spanning four decades of mobility D and the full attention range λ, the radicalized region at high λ is independent of D: geography sets the social outcome only for populations whose attention it still owns. Second, under opinion-independent Brownian mobility, geography never becomes a predictor of opinion at *any* mobility—a null result that follows from translation invariance and that qualifies a common intuition about local echo chambers. Geographic opinion structure requires opinion–position coupling: adding a Schelling-type homophilic drift of strength χ [12, 13], we find that spatial opinion domains emerge only above a Péclet-number threshold $\chi\ell/D \sim 1$, where the homophilic drift beats diffusion over the interaction range ℓ.

The ingredients of the model each have a literature: bounded confidence [7, 8, 14–16], mobile agents [9], coevolving networks [11], multiplex opinion formation [10], algorithmic bias and link recommendation [1, 2, 17], activity-driven echo chambers [3], stubborn agents [18], adaptive confidence [19], and coupled opinion–position dynamics [12, 13]. The contribution here is the combination that isolates the interaction between platform design and influence psychology under an attention constraint, and the analytical reduction that makes the resulting inversion predictable rather than merely observable.

2 Model

2.1 State variables

Each of N agents carries a position $\mathbf{r}_i(t) \in [0, L)^2$ with periodic boundary conditions and an opinion $x_i(t) \in [-1, 1]$. Agent i may also carry an influence strength s_i and a stubbornness flag. The digital layer is a directed graph $A_{ij}(t) \in \{0, 1\}$, where $A_{ij} = 1$ means that the platform currently shows content by agent j to agent i. Each agent has a fixed number k of *attention slots*, $\sum_j A_{ij} = k$ for all i and t: attention is conserved, and new digital connections can only be acquired by dropping old ones.

2.2 Dynamics

Positions follow

$$\mathrm{d}\mathbf{r}_i = \chi\, \mathbf{f}_i\, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}\mathbf{W}_i(t), \tag{1}$$

where D is the diffusion coefficient and $\mathbf{f}_i$ is an optional homophilic drift (Level 5 below; $\chi = 0$ elsewhere): the kernel-weighted mean of unit vectors pointing toward compatible neighbours ($|x_j - x_i| < \epsilon$) and away from incompatible ones. Opinions obey the Itô equation

$$\mathrm{d}x_i = \alpha_i \frac{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)\, (x_j - x_i)}{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)}\, \mathrm{d}t \;+\; \beta_i \frac{\sum_j A_{ij}\, s_j\, F(x_i, x_j)}{\sum_j A_{ij}\, s_j}\, \mathrm{d}t \;+\; \sigma_x\, \mathrm{d}B_i(t), \tag{2}$$

clipped to $[-1, 1]$, where r_{ij} is the minimum-image distance. The physical kernel is a truncated Gaussian,

$$K(r) = e^{-r^2/2\ell^2}\, \mathbf{1}(r < 3\ell), \tag{3}$$

and $B_\epsilon(\Delta) = \mathbf{1}(|\Delta| < \epsilon)$ implements bounded confidence. The truncation is not cosmetic: because the drift is normalised, an untruncated Gaussian would let an agent with no nearby compatible peers average with the entire population at $O(1)$ rate through the exponential tail, silently removing the mobility scale from the problem.

The influence function has the assimilation–indifference–repulsion form

$$F(x_i, x_j) = \begin{cases} x_j - x_i, & |x_j - x_i| < \epsilon_1, \\ 0, & \epsilon_1 \le |x_j - x_i| < \epsilon_2, \\ -\eta\, (x_j - x_i), & |x_j - x_i| \ge \epsilon_2, \end{cases} \tag{4}$$

with the repulsive branch ($\eta > 0$) active at Levels 4–5; at Levels 2–3, F reduces to bounded confidence with threshold $\epsilon_1 = \epsilon$. Empirical support for repulsive influence is discussed by Tang et al. [6].

2.3 Attention budget

The rates in Eq. (2) satisfy

$$\alpha_i = \alpha_{\text{tot}}\, (1 - \lambda), \qquad \beta_i = \alpha_{\text{tot}}\, \lambda, \qquad \lambda \in [0, 1], \tag{5}$$

so the digital attention share λ measures how much of a finite information-processing capacity is devoted to the platform. Increasing digital exposure *displaces* physical influence rather than adding to it, alongside the hard slot constraint $\sum_j A_{ij} = k$.

2.4 The platform: engagement-driven rewiring

In each interval $\mathrm{d}t$, agent i replaces one uniformly chosen attention slot with probability $\rho\,\mathrm{d}t$; the platform selects the replacement source j with probability

$$P(i \to j) \;\propto\; E(|x_i - x_j|)\; s_j, \tag{6}$$

where E is the platform's *engagement kernel*. We compare three designs:

$$E_{\mathrm{sim}}(\Delta) = e^{-\gamma\Delta}, \qquad E_{\mathrm{neu}}(\Delta) = 1, \qquad E_{\mathrm{con}}(\Delta) = e^{-(\Delta-\delta)^2/2s^2}. \tag{7}$$

The similarity kernel models algorithmic homophily with strength γ [1, 2]; the neutral kernel is an opinion-blind baseline; the controversy kernel encodes a platform that maximises engagement by promoting content at a preferred ideological distance δ, close in spirit to the post-distribution algorithms of de Arruda et al. [17].

2.5 Model levels, heterogeneity, and observables

The model is built in five nested levels: (1) Brownian motion + bounded confidence; (2) static opinion-neutral digital links; (3) adaptive similarity-driven rewiring; (4) general engagement kernels + repulsion + heavy-tailed influence strengths (s_i Pareto with exponent κ, unit mean; optional stubborn agents [18]); (5) homophilic mobility $\chi > 0$.

We measure: polarization $P = \mathrm{Var}(x)$; extremism $\langle |x| \rangle$; the number of opinion clusters n_c (gaps larger than 0.05 in sorted opinion space, groups of at least two agents); a categorical state (consensus / polarization / fragmentation for $n_c \le 1$ / $= 2$ / ≥ 3); the spatial–opinion correlation via Moran's I and via the *local agreement gap* $\Pr(|x_i - x_j| < \epsilon \mid r_{ij} < 3\ell) - \Pr(|x_i - x_j| < \epsilon)$; opinion assortativity ρ of the digital graph; modularity Q of the digital graph with respect to opinion clusters; and mean local disagreement $\langle |x_i - \bar{x}_{\partial i}| \rangle$ between an agent and the average opinion it is shown.

2.6 Numerical integration

Equations (1)–(2) are integrated by Euler–Maruyama. Unless stated otherwise, $N = 200$, $L = 1$, $\mathrm{d}t = 0.02$, $\alpha_{\mathrm{tot}} = 1$, $\epsilon = \epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$, $\ell = 0.02$, $k = 10$, $\rho = 5$, $\gamma = 4$, $\delta = 0.8$, $s = 0.2$, $\kappa = 2.5$, $\sigma_x = 0.02$, and runs last $T = 60$–80 time units. In system-size scans the box is rescaled as $L = \sqrt{N/200}$ so that density, kernel range and digital degree stay constant. With $\ell = 0.02$ the mean physical contact degree is $N\pi(3\ell)^2/L^2 \approx 2.3$, below the percolation threshold of the random geometric graph, so the instantaneous contact network is spatially fragmented and locality is meaningful. Statistics use 6–12 realisations per point (3 for the phase diagram).

3 Theory: two-bloc reduction and mean-field limit

3.1 Late stage: cross-bloc attention controls radicalization

Consider the late-stage configuration observed at Level 4: two equal blocs at opinions $\pm y$. Under fast rewiring ($\rho \gg \alpha_{\mathrm{tot}}$), the stationary probability that an attention slot of a $+$-bloc agent points into the $-$ bloc follows from Eq. (6):

$$p(y) \;=\; \frac{E(2y)}{E(0) + E(2y)}. \tag{8}$$

Within-bloc content exerts no drift ($F = 0$ at $\Delta = 0$), and cross-bloc content is repulsive once $2y \geq \epsilon_2$, so the bloc positions obey

$$\frac{\mathrm{d}y}{\mathrm{d}t} \;=\; 2\,\lambda\,\eta\,p(y)\,y, \qquad 2y \geq \epsilon_2, \tag{9}$$

with no physical restoring force: once the blocs are separated beyond the confidence bound, the physical layer only pulls agents toward their own bloc's mean. Equation (9) makes the platform ordering a statement about the shape of $p(y)$:

- **Neutral:** $p \equiv 1/2$. The drift $\lambda\eta y$ never shuts off; opinions are driven to the boundary and pinned at ± 1.
- **Controversy-driven:** $p(y) \approx 1$ while $2y \approx \delta$ (cross-bloc content is exactly what engages), so separation is fast; but as $2y \to 2$ both $E(2y)$ and $E(0)$ are far from the engagement peak and $p \to E(2)/E(0) = e^{-(2-\delta)^2/2s^2+\delta^2/2s^2}$, which for $\delta < 1$ is exponentially small. The dynamics is *kinetically arrested* just short of the boundary—the platform stops pushing once the blocs are so far apart that neither identical nor opposed content engages.
- **Similarity-driven:** $p(y) = e^{-2\gamma y}/(1 + e^{-2\gamma y})$ is small for $\gamma y \gtrsim 1$. Radicalization proceeds, but at a rate suppressed by $e^{-2\gamma y}$: a slow outward creep fed by the few cross-cutting links that stochastic rewiring keeps producing.

These three behaviours are exactly what the simulations show (Sec. 4.3): pinning at ± 1 for neutral ($\langle |x| \rangle \approx 1.00$), arrest slightly short of the boundary for controversy (≈ 0.95), and slow partial radicalization for similarity (≈ 0.81 at $T = 80$). Note also that Eq. (9) has no threshold in λ: any nonzero digital attention radicalizes eventually. What varies across designs—by orders of magnitude—is the rate, which we quantify next.

3.2 Onset: radicalization is rate-limited, not threshold-limited

One might expect a critical attention share below which the physical layer's assimilative pull protects the population. A linear stability analysis of the *uniform* state indeed predicts such thresholds (balancing the inward pull $(1-\lambda)\epsilon/2$ on edge agents against the engagement-weighted outward push, one obtains $\lambda^* \approx 0.34$ for the neutral and 0.55 for the controversy kernel). The simulations falsify this prediction: radicalization onsets already at $\lambda \approx 0.05$ (Sec. 4.5). The reason is that the uniform state never gets to apply the analysis. Bounded-confidence fragmentation forms first, on the fast timescale $\sim (1-\lambda)^{-1}$, producing blocs near $\pm y_0$ with $y_0 \approx 0.6$—and since $2y_0 > \epsilon_2$, the fragmented state is unstable to repulsion for *any* $\lambda > 0$: once separated beyond the confidence bound, the physical layer exerts no restoring force between blocs (Sec. 3.1).

What remains is a rate. Integrating Eq. (9) with $p(y) \approx p_0 = p(y_0)$ gives the radicalization time

$$t_{\mathrm{rad}}(\lambda) \;=\; \frac{\ln(y_f/y_0)}{2\,\lambda\,\eta\,p_0}, \tag{10}$$

and the apparent onset in any finite observation window T is the crossover $t_{\mathrm{rad}} = T$:

$$\lambda_c(T) \;=\; \frac{\ln(y_f/y_0)}{2\,\eta\,p_0\,T}. \tag{11}$$

Table 1 evaluates Eqs. (10)–(11) at the reference parameters with no fitting. The cross-bloc slot fraction p_0 spans two orders of magnitude across the designs, and with it the crossover: for the neutral and controversy platforms $\lambda_c \approx 0.01$–0.02—any noticeable digital attention radicalizes within the horizon—while for the similarity platform $\lambda_c \approx 1$: even full digital attention only partially radicalizes the population in the same window, which is exactly the smooth sub-envelope growth observed in the simulations. "Protection" by homophilic curation is therefore kinetic, not thermodynamic: it multiplies the time to reach the extremes by $p_0^{-1} \sim e^{2\gamma y_0}$, without changing the destination.

Table 1: Two-bloc rate analysis, Eqs. (8) and (10)–(11), at the reference parameters ($y_0 = 0.6$, $y_f = 1$, $\eta = 0.4$, $\gamma = 4$, $\delta = 0.8$, $s = 0.2$, $T = 80$). No parameters are fitted.

Platform	p_0	$t_{\text{rad}}(\lambda)$	$\lambda_c(T{=}80)$
Similarity-driven	0.008	$78.2/\lambda$	0.98
Neutral	0.500	$1.28/\lambda$	0.016
Controversy-driven	0.998	$0.64/\lambda$	0.008

3.3 Mean-field limit

In the fast-rewiring, well-mixed, $N \to \infty$ limit, the opinion marginal obeys a McKean–Vlasov dynamics: each agent feels the global bounded-confidence mean with weight $1-\lambda$ and an engagement-weighted influence average with weight λ,

$$\mathrm{d}x = (1-\lambda)\,\frac{\mathbb{E}\big[(x'-x)\,\mathbf{1}(|x'-x|<\epsilon)\big]}{\mathbb{P}\big(|x'-x|<\epsilon\big)}\,\mathrm{d}t + \lambda\,\frac{\mathbb{E}\big[s'\,E(|x'-x|)\,F(x,x')\big]}{\mathbb{E}\big[s'\,E(|x'-x|)\big]}\,\mathrm{d}t + \sigma_x\,\mathrm{d}B, \tag{12}$$

where (x', s') is an independent draw from the population law. We integrate Eq. (12) as a spaceless interacting-particle system and compare it with the spatial agent-based model in Sec. 4.5: it reproduces the full $\mathrm{Var}(x)$–λ curves of all three platforms with no fitted parameters, confirming that the inversion is a bulk property of the opinion dynamics, not an artefact of space, finite degree, or slow rewiring.

4 Results

4.1 Baselines: mobility and neutral connectivity (Levels 1–2)

Figures 1–3 establish the assimilative baselines. In the purely physical model, the competition between the local convergence time $\tau_{\text{op}} \sim \alpha_{\text{tot}}^{-1}$ and the neighbourhood-renewal time $\tau_\ell \sim \ell^2/D$ controls how many opinion clusters survive: $n_c \approx 5$–9 frozen local clusters for $D \lesssim 10^{-4}$, merging into the mean-field cluster set ($n_c \approx 2$–3 at $\epsilon = 0.3$) for $D \gtrsim 10^{-2}$, consistent with mobile-agent bounded-confidence simulations [7, 9]. Adding a *static, opinion-blind* digital layer at low mobility heals this excess fragmentation: a modest attention share ($\lambda \approx 0.2$) already merges the frozen local clusters into the mean-field set, beyond which more digital attention changes nothing (Fig. 3). Opinion-blind long-range exposure acts like infinite-range mobility [10]. In the assimilative world, connectivity is benign; everything that follows is about what curation and repulsion do to this baseline.

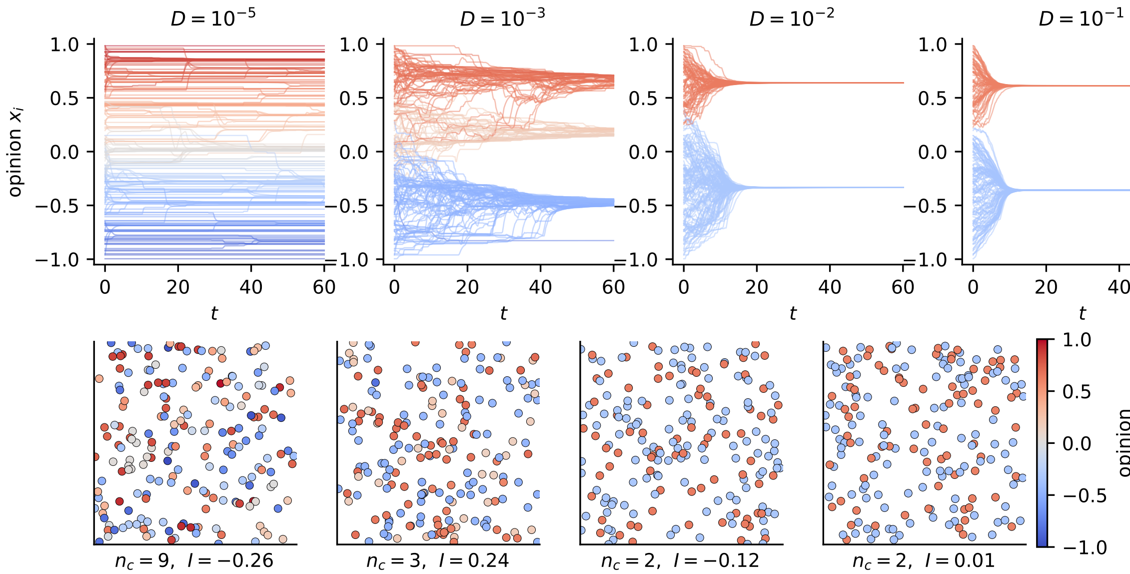

Figure 1: Purely physical dynamics (Level 1). Top: opinion trajectories $x_i(t)$ for increasing diffusion coefficient D; colours encode final opinions. Bottom: final spatial configurations. Panel captions report the cluster count n_c and Moran's I. Low mobility freezes many coexisting clusters; high mobility recovers the mean-field bounded-confidence outcome. Note the absence of spatial colour domains at any D.

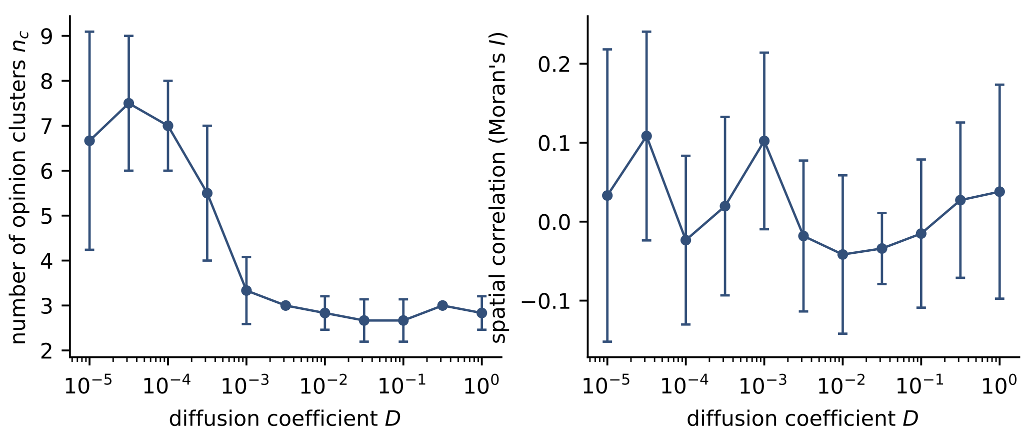

Figure 2: Level-1 sweep over the diffusion coefficient. Left: mean number of surviving opinion clusters. Right: Moran's I between opinion and position (mean $\pm$ s.d. over 6 realisations). Mobility interpolates between local fragmentation and mean-field behaviour while the spatial–opinion correlation stays at noise level throughout.

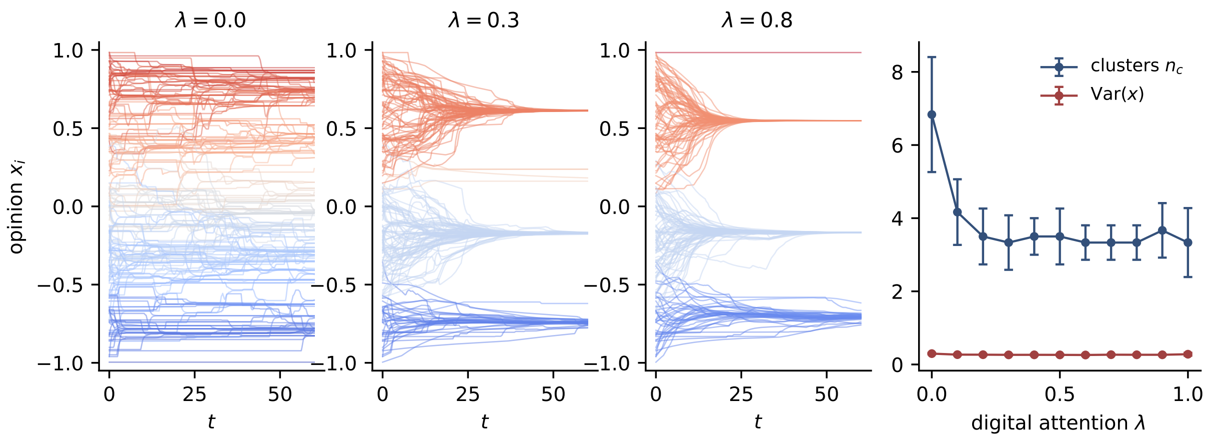

Figure 3: Level 2: static random digital layer at low mobility ($D = 10^{-4}$). Left three panels: opinion trajectories for increasing digital attention share λ. Right: cluster count and polarization versus λ (mean $\pm$ s.d.).

4.2 Algorithmic homophily builds delocalised echo chambers (Level 3)

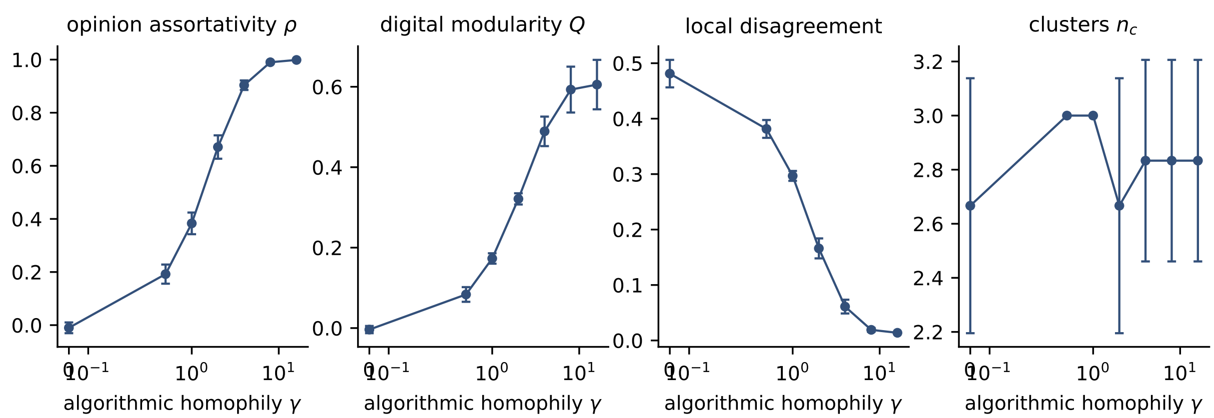

Figure 4: Level 3: adaptive digital layer with similarity-driven recommendation, $\lambda = 0.5$, $D = 10^{-3}$. From left to right: opinion assortativity of the digital graph, modularity with respect to opinion clusters, mean local disagreement, and cluster count as functions of the algorithmic homophily γ (mean $\pm$ s.d. over 6 runs).

With similarity-driven rewiring (still purely assimilative influence), the feedback loop opinion $\to$ topology $\to$ exposure $\to$ opinion closes as γ grows (Fig. 4): opinion assortativity climbs to one, digital modularity rises, and the disagreement visible in an agent's feed collapses while global fragmentation persists. This is an echo chamber in the precise sense of Baumann et al. [3]—high global variance, near-zero visible disagreement—via the mechanism of Sîrbu et al. [1] and Santos et al. [2] transplanted into a coevolving multiplex [11]. The communities are spatially delocalised: geography carries no trace of them. In the assimilative world this is the worst a platform can do in our model: freeze moderate fragmentation and hide it from its participants.

4.3 The inversion: uncurated exposure radicalizes (Level 4)

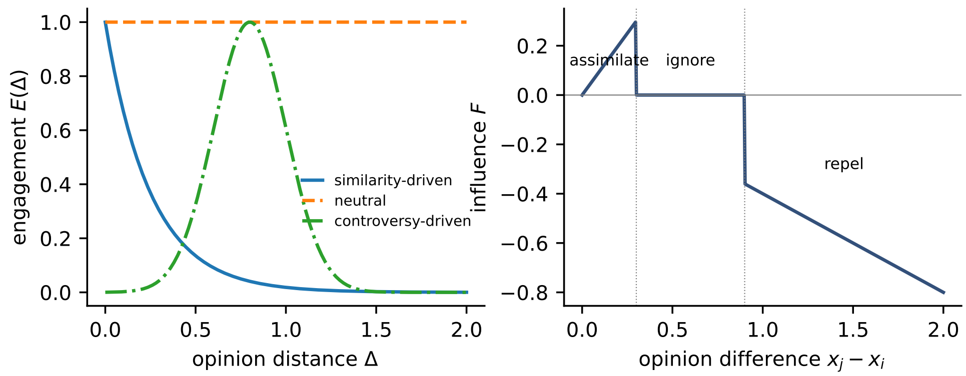

Figure 5: Level-4 ingredients. Left: the three engagement kernels of Eq. (7) ($\gamma = 4$, $\delta = 0.8$, $s = 0.2$). Right: the assimilation–indifference–repulsion influence function of Eq. (4) ($\epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$).

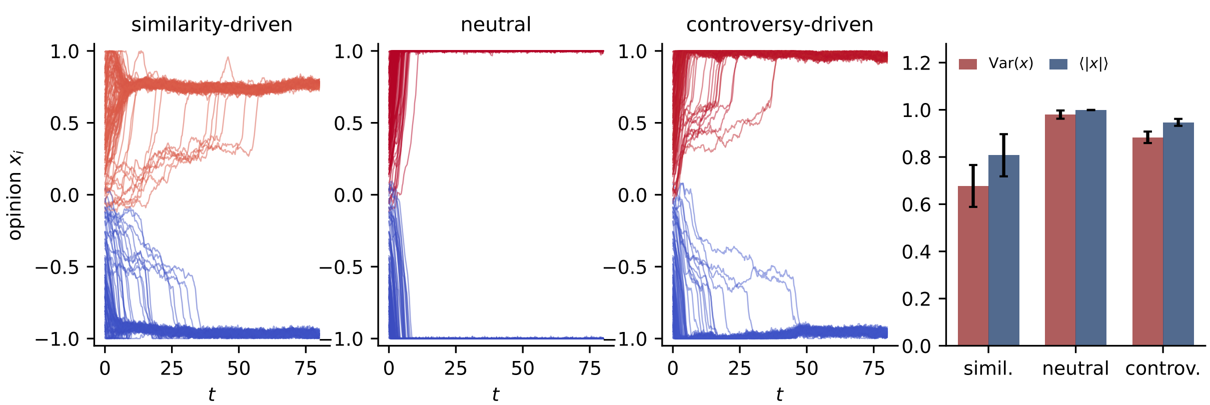

Figure 6: Level 4: platform designs compared under repulsive influence and heavy-tailed influence strengths ($\lambda = 0.6$). Left three panels: representative opinion trajectories. Right: polarization $\mathrm{Var}(x)$ and extremism $\langle |x| \rangle$ (mean $\pm$ s.d. over 8 runs).

Activating the repulsive branch and heavy-tailed influence strengths (Fig. 5) produces the inversion (Fig. 6). All three designs end with two antagonistic blocs, but with sharply different degree and speed, in the order predicted by the two-bloc reduction of Sec. 3.1. The neutral platform is the most polarizing ($\mathrm{Var}(x) = 0.98 \pm 0.02$, opinions pinned at ± 1 within a few time units): with $p = 1/2$, half of every feed is repulsion-triggering at all times. The controversy platform is nearly as extreme (0.88 ± 0.02) but arrests just short of the boundary ($\langle |x| \rangle \approx 0.95$), as predicted by the collapse of $p(y)$ when $2y \to 2$. The similarity platform is the least polarizing (0.68 ± 0.09): hiding opposed content starves the repulsive channel, leaving only the slow creep fed by residual cross-cutting links. The network signatures decouple from the opinion signatures: the neutral platform reaches maximal polarization with *zero* opinion assortativity ($\rho \approx 0$)—polarization without echo chambers—whereas both curated platforms end almost perfectly assortative ($\rho \approx 0.9$–1.0).

4.4 Robustness: uncurated exposure is the envelope

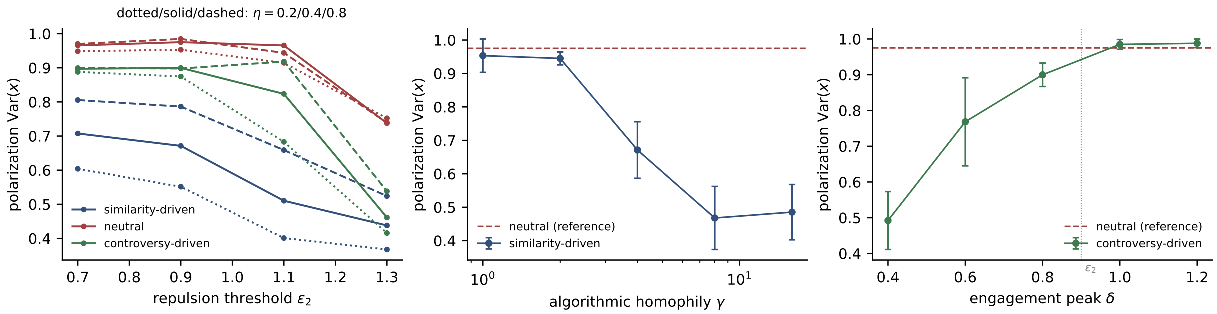

Figure 7: Robustness of the inversion. Left: final polarization versus the repulsion threshold ϵ_2 for $\eta = 0.2/0.4/0.8$ (dotted/solid/dashed); colours denote platforms (blue similarity, red neutral, green controversy). Centre: similarity platform versus algorithmic homophily γ, with the neutral platform as reference. Right: controversy platform versus engagement peak δ; the dotted line marks $\delta = \epsilon_2$. Mean $\pm$ s.d. over 6 runs.

Figure 7 tests the ordering across the repulsion and curation parameters. Across the full (η, ϵ_2) grid the order neutral $\geq$ controversy $>$ similarity holds uniformly; all platforms polarize less as ϵ_2 grows (repulsion becomes harder to trigger) and more as η grows. The two curated platforms interpolate toward the neutral level in exactly the way the theory predicts. The similarity platform approaches the neutral bound as $\gamma \to 1$ (curation too weak to withhold repulsive content) and its protective effect saturates for $\gamma \gtrsim 8$. The controversy platform sits below the neutral bound while its engagement peak $\delta < \epsilon_2$ (much of its exposure lands in the indifference band) and reaches the bound once $\delta \geq \epsilon_2$, when engagement maximisation and repulsion maximisation coincide. Uncurated exposure is thus the *upper envelope* of radicalization: a platform is exactly as dangerous as the amount of repulsive-zone exposure it fails to withhold.

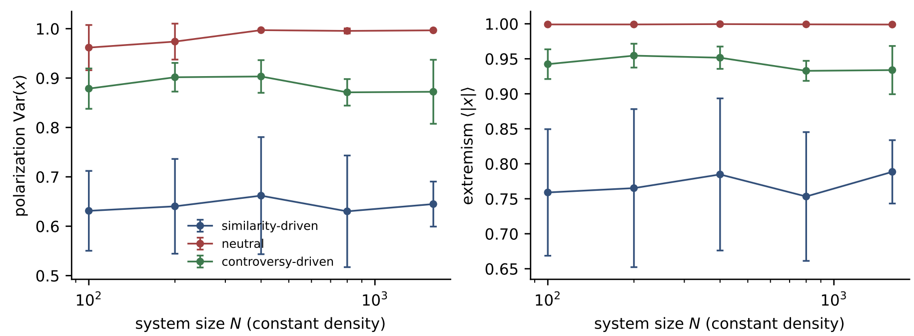

Figure 8: System-size dependence of the inversion at constant density ($L = \sqrt{N/200}$), $\lambda = 0.6$. Polarization (left) and extremism (right) for the three platforms, $N = 100$–1600.

Figure 8 scans the system size at constant density up to $N = 1600$. The three platforms' polarization and extremism are essentially flat in N: the inversion is a bulk property, not a finite-size artefact.

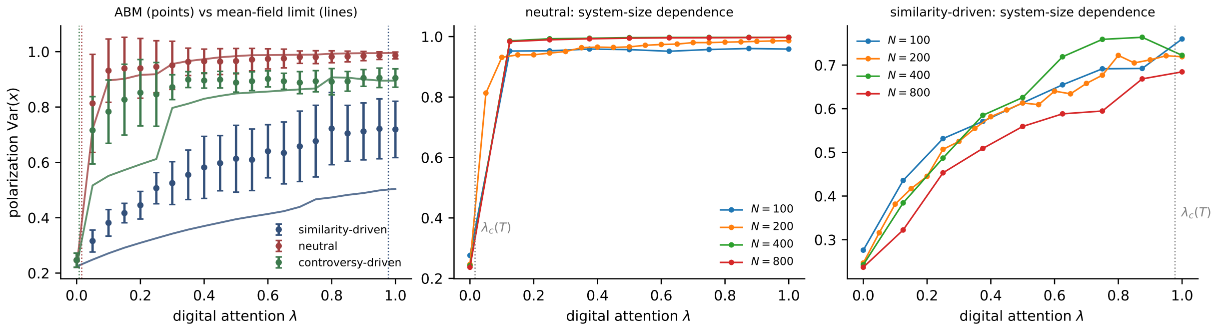

Figure 9: Radicalization onset in the digital attention share. Left: final polarization versus λ for the three platforms; points are the spatial ABM ($N = 200$, 12 seeds), solid lines the McKean–Vlasov mean-field limit of Eq. (12), dotted verticals the parameter-free finite-horizon crossovers $\lambda_c(T)$ of Table 1. Centre and right: system-size dependence of the $\mathrm{Var}(x)$–λ curves for the neutral and similarity platforms.

4.5 The radicalization threshold and the mean-field limit

Figure 9 compares the simulated $\mathrm{Var}(x)$–λ curves with the theory of Sec. 3. The observations match the rate picture in every particular. The neutral and controversy platforms jump to their saturated values already at $\lambda \approx 0.05$, consistent with $\lambda_c \approx 0.01$–0.02 and inconsistent with the naive uniform-state thresholds (0.34, 0.55); their curves are essentially independent of system size (centre panel), as expected for a finite-horizon crossover rather than a phase transition. The similarity platform grows smoothly along λ toward a partial radicalization, consistent with $t_{\mathrm{rad}} \approx 78/\lambda$ against the horizon $T = 80$; its weak downward drift with N (right panel) reflects the averaging-out of finite-attention fluctuations—with only $k = 10$ slots, occasional high-influence cross-bloc sources produce bursts of repulsion that the $N \to \infty$, fast-rewiring limit smooths away, which is also why the mean-field curve (solid blue) sits below the ABM points while tracking their shape. The mean-field integration reproduces the neutral and controversy curves nearly quantitatively with no fitted parameters.

4.6 Phase diagram: attention neutralises geography

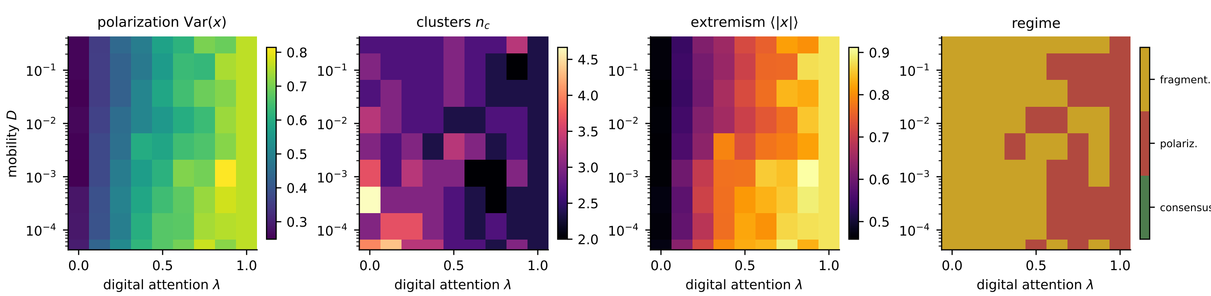

Figure 10: Phase diagram of the full Level-4 model (similarity-driven engagement, repulsion, heavy-tailed influence) in the mobility–digital-attention plane. From left to right: polarization $\mathrm{Var}(x)$, cluster count n_c, extremism $\langle |x| \rangle$, and the majority categorical state over 3 realisations per grid point.

Figure 10 assembles the global picture in the (D, λ) plane. At $\lambda \approx 0$ the physical layer decides the outcome and the Level-1 crossover is recovered: frozen local fragmentation at low D, the mean-field cluster set at high D. As λ grows the platform captures the attention budget, the repulsive channel activates through the residual cross-cutting exposure, and both polarization and extremism rise across *all* mobilities: the high-λ region is a bimodal, radicalised state essentially independent of D. Once the platform curates most of an agent's exposure, it no longer matters how fast people move: geography sets the outcome only for societies whose attention it still owns.

4.7 Geography: a null result and its repair (Levels 1 and 5)

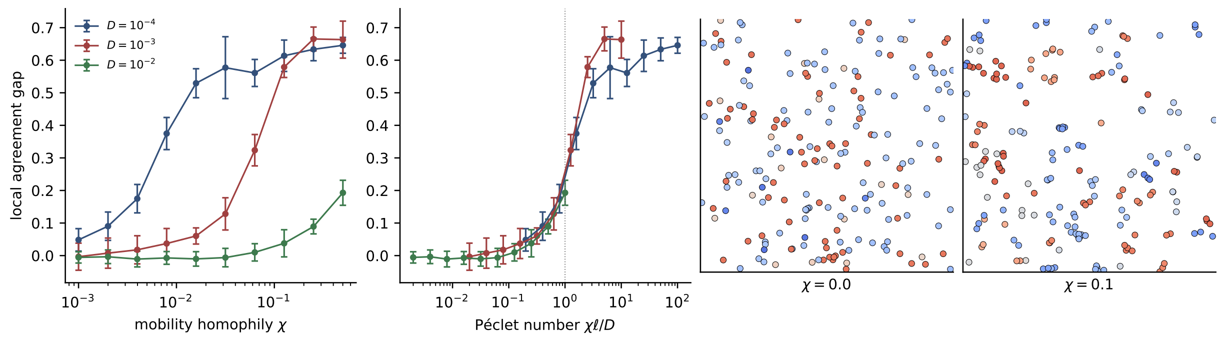

Figure 11: Level 5: homophilic mobility restores geographic opinion structure. Left: local agreement gap versus the homophilic drift strength χ for three mobilities. Centre-left: the same data against the Péclet number $\chi\ell/D$; the curves approximately collapse, with onset at $\mathrm{Pe} \sim 1$ (dotted). Right: final spatial configurations at $D = 10^{-3}$ without $(\chi = 0)$ and with $(\chi = 0.1)$ homophilic mobility.

The second panel of Fig. 2 contains a null result that qualifies a common intuition. Moran's I—and the local agreement gap—never departs from noise level, *even in the frozen regime where interactions are strictly local*. The first-order part of this statement is a symmetry: on the torus, the initial law and the dynamics with $\chi = 0$ are translation invariant, so $\mathbb{E}[x_i \mid \mathbf{r}_i = \mathbf{r}]$ is independent of $\mathbf{r}$ at all times—no deterministic opinion map can exist. Symmetry does not forbid *pair*-level structure (nearby agents interacted more), but the measured pair correlations also stay at noise level: disagreeing agents interleave at the same locations, interacting with disjoint compatible subsets, because nothing couples opinion to position. Locality controls *how many* opinions survive but not *where* they live. Claims that low mobility alone produces geographic echo chambers therefore require an opinion–position coupling, however weak.

Level 5 quantifies how much coupling is needed. With the homophilic drift of Eq. (1) ($\chi > 0$: approach compatible neighbours, avoid incompatible ones [12, 13]), spatial opinion domains appear (Fig. 11). The control parameter is the Péclet number comparing drift and diffusion over the interaction range,

$$\mathrm{Pe} = \frac{\chi\,\ell}{D}, \tag{13}$$

and the local agreement gap for different mobilities approximately collapses onto a single curve in Pe, rising from noise level at $\mathrm{Pe} \lesssim 0.3$ to strong spatial segregation for $\mathrm{Pe} \gtrsim 3$. Geographic opinion structure is thus a threshold phenomenon in the strength of opinion-dependent mobility—and the Brownian model of Levels 1–4 sits below the threshold by construction, making it the correct null baseline.

5 Discussion

Summary. We combined bounded-confidence opinion dynamics, Brownian and homophilic mobility, and an algorithmically curated digital layer under a finite attention budget, and found that the interaction between platform design and influence psychology—not either alone—decides the collective outcome. With assimilative influence, the neutral platform is a defragmenting force and homophilic curation builds echo chambers; with a repulsive channel, every design radicalizes, the neutral platform is the upper envelope, and homophilic curation is a buffer. The two-bloc reduction, the parameter-free finite-horizon crossover $\lambda_c(T)$, and the McKean–Vlasov limit make these statements analytical rather than merely observational, and the robustness and finite-size scans show they are bulk properties of the model. A cautionary methodological note falls out for free: the naïve linear-stability analysis of the uniform state predicts thresholds an order of magnitude too large, because fragmentation preempts it—in coupled opinion–network systems, the state that matters for stability is the one the fast dynamics builds first.

Relation to experiments. The inversion offers a mechanistic, population-scale account of the field experiment of Bail et al. [4], in which a month of curated exposure to opposing views made Republican participants substantially more conservative: in our terms, an intervention that raises cross-cutting exposure moves a similarity-curated population toward the neutral envelope, and when the repulsive channel dominates—the micro-mechanism with the stronger empirical support in longitudinal network data [6]—that movement increases polarization. The model also predicts the conditions under which the same intervention helps: when influence is assimilative (weak prior commitments, moderate opinion distances), added exposure diversity heals fragmentation (Sec. 4.1). Interventions on recommendation algorithms that ignore which influence regime the population is in can therefore backfire in either direction.

The geography null result. Under opinion-independent mobility there is no geographic opinion structure to destroy: position and opinion stay uncorrelated at every mobility, by symmetry at first order and empirically at pair level. The geographic clustering of opinion observed in real societies must then be produced by opinion–position coupling—homophilic migration, socially structured mobility, or spatially correlated exogenous influence—and Level 5 shows the coupling need only exceed a modest Péclet threshold $\chi\ell/D \sim 1$. This sharpens, rather than contradicts, the mobile-agent literature [9, 12]: what matters is not whether agents move, but whether their movement knows about opinions.

Limitations and outlook. Opinions are one-dimensional and confidence bounds static; adaptive-confidence extensions are natural [19]. The two-bloc reduction is heuristic—it assumes symmetric blocs and fast rewiring—and its quantitative success invites a rigorous analysis of the McKean–Vlasov equation (12), including the metastability structure behind the finite-horizon crossover [15, 16]. The attention budget is a fixed split rather than a dynamic allocation; letting λ itself be optimised by an engagement-maximising platform closes an obvious loop. Stubborn agents are implemented but unexplored; sweeping their number and placement against platform designs connects to opinion control [18]. Finally, the model's sharpest empirical claim—that exposure diversity helps or harms depending on the prevalence of negative influence—is testable in principle by conditioning field-experiment outcomes on measured latitudes of acceptance.

Reproducibility

All code, figures, and animation scripts are contained in the accompanying repository. Each figure is produced by a single script under `scripts/` (the theory curves by `scripts/theory_twobloc.py`); the model implementation lives in `src/socialsim/`; Manim animations of the coupled dynamics are in `visualizations/`.

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